

Radar Telemetry Over LoRa

Project Progress Update
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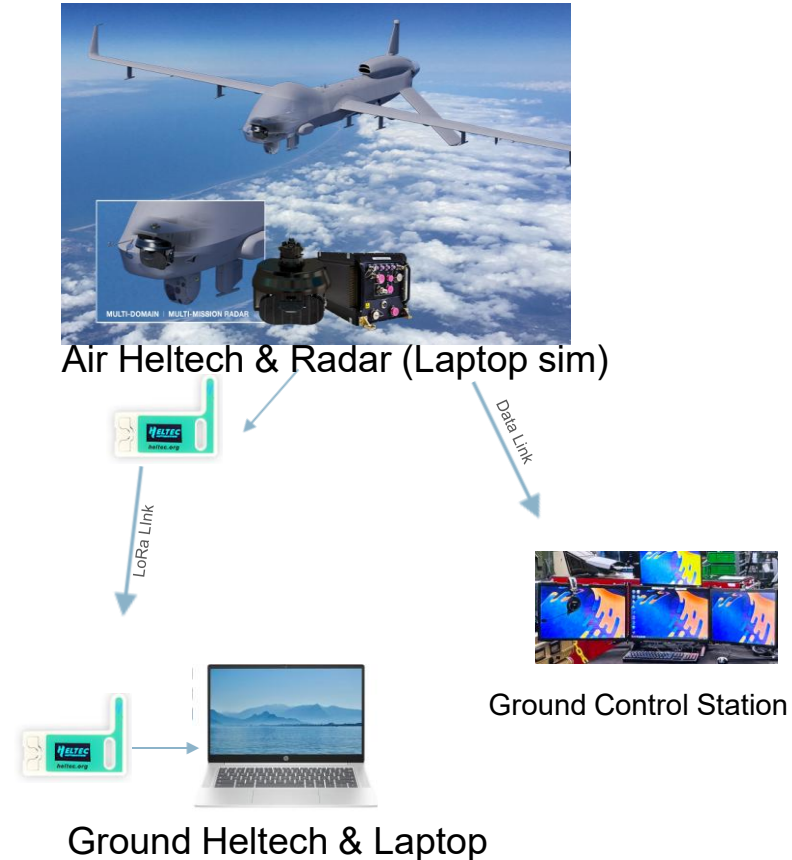
Project Goal

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- Build a wireless radar test link between an air side and a ground side
- Let the ground side send commands
- Let the air side return updated radar status
- Will help determine whether a problem is in the radar or in the ground control station network

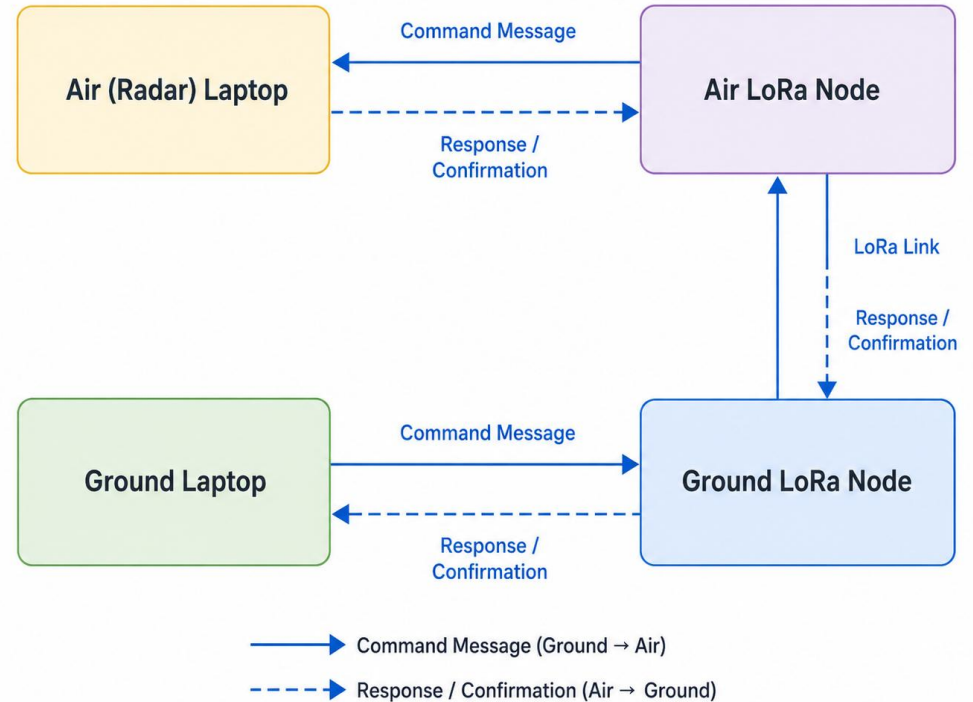
Simple Explanation

- The ground laptop sends a command
- The air laptop receives it
- The air laptop sends the updated radar status back
- Both sides show the data on screen



System Setup

- 2 Laptops
- 2 Heltec LoRa boards
- 2 Heltec LoRa boards
- Ground Heltec sends commands over LoRa
- Air Heltec forwards commands to the air laptop
- Air laptop sends updated radar status back
- Ground laptop shows and logs the returned status



Modes Being Tested & Test Card

Radar Modes

- Motion Target Indicator (MTI)
- SARVideo
- Maritime Large Vessel
- Maritime Small Vessel
- TX Enable
- TX Disable

Test Card

- Send commands from the ground laptop
- Check that the air laptop updates correctly
- Check that the ground laptop receives the correct reply
- Record success or failure for each test
- Static test from 0ft to 200ft at increments of 20 feet.
- Slow speed (10 to 20 MPH) driving test

Ground HMI

Ground Radar Telemetry HMI

Link Status

Link: OK Last Update: 2026-05-07 20:24:11 Packet Type: Status CID: 0 Result: MISMATCH

Log File: ground_log_20260507_202342.csv

Commanded State

Mode: MTI

TX State: Enabled

MTI

SARVideo

Maritime Large

Maritime Small

TX Enable

TX Disable

Resend Command

Last Command Packet:

Reported Radar State

Mode: SARVideo

TX State: Disabled

Health: OK

Alarm: 0

RSSI (dBm): -33.00

SNR (dB): 12.00

Raw Packet: T=STAT,CID=0,MODE=SARV,TX=DIS,H=OK,A=0,RSSI=-33.00,SNR=12.00

Test Metrics

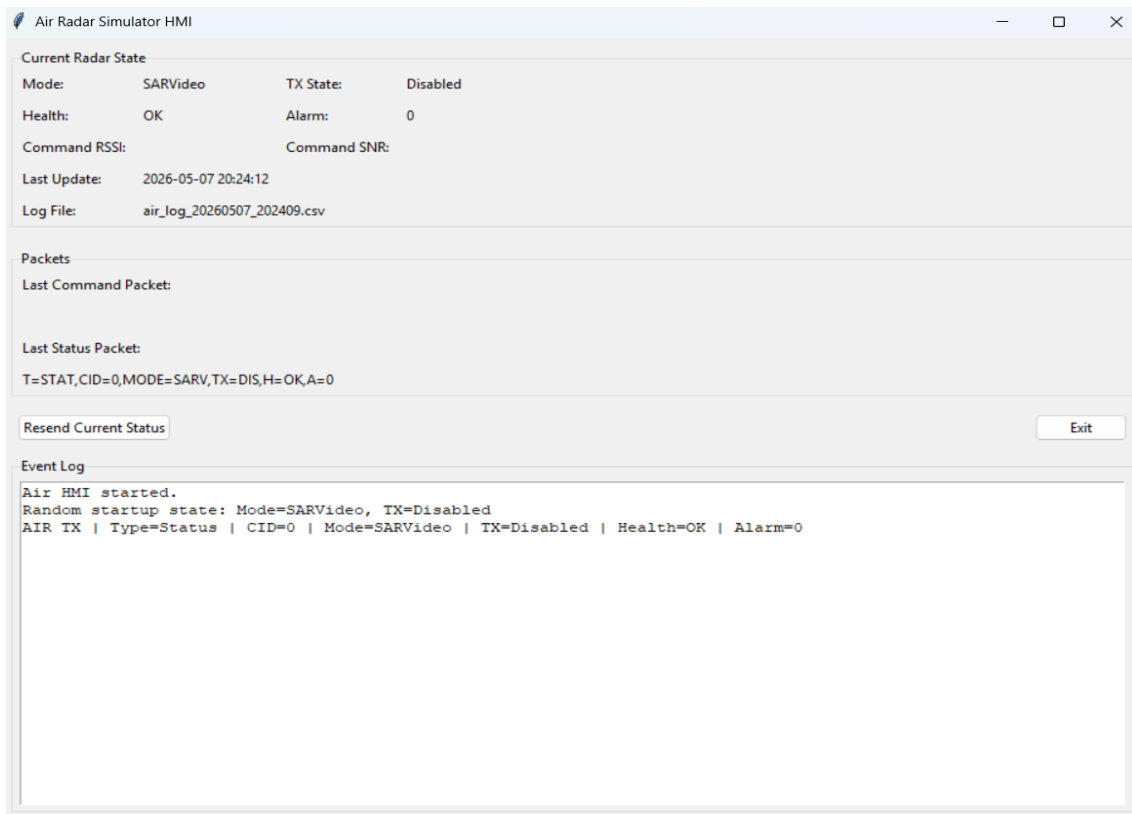
Total Commands: 0 Total Responses: 1 Mode Changes: 0 TX Changes: 0 Matches: 0 Mismatches: 0

Exit

Event Log

Ground HMI started.
GROUND RX | Type=Status | CID=0 | Mode=SARVideo | TX=Disabled | Health=OK | Alarm=0 | RSSI=-33.00 dBm | SNR=12.00 dB | Latency=s | Match=NO

Air Radar HMI



Drive Test

Method

- Speed: 10–15 mph
- Route: around the block
- Trials: 2
- Check: command and returned status match

Results

- 100% match to 60 yards
- Failed beyond 60 yards and lost packets
- Same result in both trials
 - 17 Mode changes
 - 6 total response
 - 11 total lost packets

Drive Test Results from Ground and Air HMI

Ground Radar Telemetry HMI

Link Status

Link: STALE Last Update: 2026-05-07 20:34:41 Packet Type: Status CID: 17 Result: MATCH

Log File: ground_log_20260507_202342.csv

Commanded State

Mode: SARVideo

TX State: Enabled

MTI

SARVideo

Maritime Large

Maritime Small

TX Enable

TX Disable

Resend Command

Reported Radar State

Mode: SARVideo

TX State: Enabled

Health: OK

Alarm: 0

RSSI (dBm): -90.00

SNR (dB): 8.00

Raw Packet: T=STAT,CID=17,MODE=SARV,TX=EN,H=OK,A=0,RSSI=-90.00,SNR=8.00

Last Command Packet:

T=CMD,CID=17,TS=1778211280.866954,MODE=SARV,TX=EN

Test Metrics

Total Commands: 17 Total Responses: 6 Mode Changes: 10 TX Changes: 4 Matches: 5 Mismatches: 0

Event Log

GROUND TX | Type=Command | CID=7 | Mode=Maritime Small | TX=Disabled

GROUND TX | Type=Command | CID=8 | Mode=MTI | TX=Disabled

GROUND TX | Type=Command | CID=9 | Mode=Maritime Large | TX=Disabled

GROUND TX | Type=Command | CID=10 | Mode=Maritime Large | TX=Enabled

GROUND RX | Type=Status | CID=10 | Mode=Maritime Large | TX=Enabled | Health=OK | Alarm=0 | RSSI=-82.00 dBm | SNR=11.25 dB | Latency=0.3s | Match=YES

GROUND TX | Type=Command | CID=11 | Mode=SARVideo | TX=Enabled

GROUND TX | Type=Command | CID=12 | Mode=Maritime Small | TX=Enabled

GROUND TX | Type=Command | CID=13 | Mode=Maritime Small | TX=Disabled

GROUND TX | Type=Command | CID=14 | Mode=MTI | TX=Disabled

GROUND TX | Type=Command | CID=15 | Mode=Maritime Large | TX=Disabled

GROUND TX | Type=Command | CID=16 | Mode=Maritime Large | TX=Enabled

GROUND RX | Type=Status | CID=16 | Mode=Maritime Large | TX=Enabled | Health=OK | Alarm=0 | RSSI=-102.00 dBm | SNR=-3.25 dB | Latency=0.29s | Match=YE

Air Radar Simulator HMI

Current Radar State

Mode: SARVideo TX State: Enabled

Health: OK Alarm: 0

Command RSSI: -93.00 Command SNR: 1.25

Last Update: 2026-05-07 20:34:42

Log File: air_log_20260507_202409.csv

Packets

Last Command Packet:

T=CMD,CID=17,TS=1778211280.866954,MODE=SARV,TX=EN,RSSI=-93.00,SNR=1.25

Last Status Packet:

T=STAT,CID=17,MODE=SARV,TX=EN,H=OK,A=0

Resend Current Status

Exit

Event Log

Air HMI started.

Random startup state: Mode=SARVideo, TX=Disabled

AIR TX | Type=Status | CID=0 | Mode=SARVideo | TX=Disabled | Health=OK | Alarm=0

AIR RX | Type=Command | CID=1 | Mode=MTI | TX=Enabled | RSSI=-77.00 dBm | SNR=7.00 dB

AIR TX | Type=Status | CID=1 | Mode=MTI | TX=Enabled | Health=OK | Alarm=0

AIR RX | Type=Command | CID=2 | Mode=Maritime Large | TX=Enabled | RSSI=-84.00 dBm | SNR=2.75 dB

AIR TX | Type=Status | CID=2 | Mode=Maritime Large | TX=Enabled | Health=OK | Alarm=0

AIR RX | Type=Command | CID=9 | Mode=Maritime Large | TX=Disabled | RSSI=-101.00 dBm | SNR=-5.50 dB

AIR TX | Type=Status | CID=9 | Mode=Maritime Large | TX=Disabled | Health=OK | Alarm=0

AIR RX | Type=Command | CID=10 | Mode=Maritime Large | TX=Enabled | RSSI=-84.00 dBm | SNR=8.50 dB

AIR TX | Type=Status | CID=10 | Mode=Maritime Large | TX=Enabled | Health=OK | Alarm=0

AIR RX | Type=Command | CID=16 | Mode=Maritime Large | TX=Enabled | RSSI=-99.00 dBm | SNR=-4.25 dB

AIR TX | Type=Status | CID=16 | Mode=Maritime Large | TX=Enabled | Health=OK | Alarm=0

AIR RX | Type=Command | CID=17 | Mode=SARVideo | TX=Enabled | RSSI=-93.00 dBm | SNR=1.25 dB

AIR TX | Type=Status | CID=17 | Mode=SARVideo | TX=Enabled | Health=OK | Alarm=0

MVP Status and Next Steps

MVP Status

- The MVP is working
- The ground laptop can send commands to the air laptop
- The air laptop can return the updated radar status
- Both HMIs display and log the test data
- Bench testing and driving tests have been completed

Next Steps – May9th to June 5th

- Repeat the driving test and collect more data
- Compare packet success, latency, RSSI, and SNR
- Identify why the link drops after about 60 yards during motion
- Visualize trial data by generating graphs and tables from saved CSV files.
- Finish the final report, GitHub updates, and video presentation